A PROJECT TITLE

ON

"Phishing Website Detection Using Random Forest and NLP-Based URL Analysis"

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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1. Abstract:

Phishing websites remain one of the most common and damaging forms of cyberattack, tricking users into revealing sensitive credentials, financial details, and personal information through deceptive web pages that mimic legitimate services. Traditional detection approaches, such as blacklisting known malicious URLs or relying on manual heuristics, struggle to keep pace with the speed at which new phishing domains are created and often fail against previously unseen sites. This project proposes a hybrid machine learning framework that combines lexical and structural URL feature analysis with Natural Language Processing (NLP) techniques to detect phishing websites more accurately and in real time. A Random Forest classifier is trained on a labelled dataset of phishing and legitimate URLs using engineered features such as URL length, presence of special characters, domain age, use of IP addresses instead of domain names, and the frequency of suspicious keywords. In parallel, NLP-based text vectorization methods, including TF-IDF and n-gram analysis, are applied to the textual composition of the URL and, where available, page content, allowing the model to capture subtle linguistic patterns typical of phishing attempts, such as brand impersonation and misspelled domain names. The two feature sets are combined to train the Random Forest model, which is chosen for its robustness to overfitting, interpretability through feature importance ranking, and strong performance on structured, tabular data. The system is evaluated using standard metrics including accuracy, precision, recall, and F1-score, and compared against baseline classifiers to demonstrate its effectiveness. Experimental results indicate that the combined URL-feature and NLP-based approach significantly improves phishing detection accuracy compared to using either technique alone, while maintaining low false-positive rates. The proposed system offers a lightweight, scalable solution that can be integrated into browser extensions, email filters, or enterprise security gateways to proactively protect users from phishing threats.

2. INTRODUCTION:

Phishing is a form of social engineering attack in which malicious actors create fraudulent websites that closely resemble trusted platforms such as banking portals, e-commerce sites, or social media services, with the intent of stealing sensitive user information. Despite growing awareness and the deployment of security tools, phishing continues to be one of the leading causes of data breaches and financial fraud worldwide, largely because attackers constantly evolve their tactics to bypass existing defences.

Conventional anti-phishing techniques, such as blacklist-based filtering and rule-based heuristics, are inherently reactive: they can only block URLs that have already been identified and reported as malicious, leaving a critical window of vulnerability during which newly created phishing sites remain undetected. This limitation has motivated the shift towards machine learning based detection systems, which can generalize from patterns observed in known phishing and legitimate websites to identify previously unseen threats.

This project focuses on building a phishing website detection system that leverages the Random Forest algorithm, an ensemble learning method known for its high accuracy and resistance to overfitting, in conjunction with Natural Language Processing techniques applied to URL text. By analysing both the structural characteristics of a URL (such as length, special characters, and domain properties) and its linguistic patterns (such as suspicious keywords and brand-name imitation), the proposed system aims to achieve higher detection accuracy than approaches relying on a single feature type. The remainder of

this report details the dataset, feature engineering process, model design, implementation, and evaluation of the proposed system.

3. SOFTWARE REQUIREMENTS:

- **Operating System:** Windows 10 / 11 or Ubuntu 20.04 LTS or later
- **Programming Language:** Python 3.9 or above
- **IDE / Development Environment:** Jupyter Notebook / Google Colab / VS Code
- **Machine Learning Libraries:** scikit-learn, pandas, NumPy
- **NLP Libraries:** NLTK, spaCy, scikit-learn (TF-IDF, CountVectorizer)
- **Data Visualization:** Matplotlib, Seaborn
- **Web Framework (optional deployment):** Flask / Streamlit
- **Version Control:** Git and GitHub
- **Dataset Source:** UCI Machine Learning Repository / PhishTank / Kaggle phishing URL datasets



Signature of the Guide



Course Instructor